24 Landscape Legibility within a funerary landscape: a GIS-based quantitative analysis of tomb visibility and human movement in Wadi Halfayin, Ad-Dakhiliyah, Sultanate of Oman

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The placement of funerary monuments across a landscape is a process that has occurred throughout human history. In prehistoric Arabia, tombs cluster on ridges, terraces and plains. What spatial logic governs this distribution? Does a single uniform set of criteria underlie all funerary structures, or does the pattern reflect millennia of shifting social, cultural and practical norms?

This paper uses Wadi Halfayin, Ad-Dakhiliyah, Oman, and a corpus of 683 tombs spanning over three thousand years to investigate this question. The entire workflow is built on Free, Libre and Open Source (FLOS) software and open data: all spatial analysis was conducted in QGIS 3.34 (Prizren). It also used the publicly available ALOS PALSAR 12.5m DEM as the Digital Elevation Model to carry out spatial analysis. This study argues that full reproducibility through FLOS tools should be standard in computational landscape archaeology.

Viewshed and least-cost path analysis are integrated and tested against 10,000-iteration Monte Carlo simulations. Results show tombs record high cumulative visibility from movement nodes and cluster closer to least-cost paths than chance predicts but visibility along routes themselves was non-significant. This pattern of proximity without coincidence supports Kevin Lynch's (1960) legibility framework: tombs were placed to be seen from junctions, not from the paths between them. Chronological analysis reveals a shift over time, with Hafit tombs dominating visually exposed ridgelines while later periods diversify.